

1. Classical: FD solver

exact (convergent) per case
no learning, no data
cost paid on *every* solve

learn the
solution

2. PINN (7 architectures)

learns $(x, t) \mapsto u$, one case
physics loss, mesh-free, no data
hours of training per material

learn the
solver

3. FNO (operator learning)

learns material $\mapsto$ wavefield
supervised on FD data
new material = one forward pass

per-case cost decreases $\longrightarrow$

up-front cost (data, training) increases $\longrightarrow$